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Open Research Knowledge Graph: A Large-scale Neuro-Symbolic Knowledge Organization System

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Abstract. This article presents the implementation of a neuro-symbolic system within the Open Research Knowledge Graph (ORKG), a platform designed to collect and organize scientific knowledge in a structured, machine-readable format. Our approach leverages the strengths of symbolic knowledge representation to encode complex relationships and domain-specific rules, combined with the pattern recognition capabilities of neural networks to process large volumes of unstructured data, in particular scientific articles in the form of narrative text documents. With the ORKG, we developed a hybrid system that integrates a knowledge graph (symbolic system) with neural networks (subsymbolic system), enabling the machine-assisted extraction of scientific knowledge from research papers and symbolic knowledge representation, curation, organisation, and reuse. The implementation involves constructing a symbolic knowledge representation of the ORKG, training neural models on annotated datasets, and designing algorithms to synergize the two systems. Our findings suggest that neuro-symbolic integration enhances the ORKG's ability to support complex queries, infer new relationships, and provide more robust and explainable AI-driven insights. This research contributes to the broader field of AI by showcasing the practical benefits of combining neural and symbolic methodologies in knowledge management systems.

Keywords. Neuro-symbolic AI, scholarly communication, knowledge graphs, large language models

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1. Introduction

In the rapidly evolving landscape of scientific research and scholarship, the dissemination and utilization of knowledge are paramount. Traditional methods of publishing and sharing scientific knowledge, while valuable, silo knowledge within dense, static documents that challenge integration, comparison, and reuse across disciplines. The Open Research Knowledge Graph [1] (ORKG, orkg.org) presented in this chapter is a pioneering initiative that reimagines the future of scholarly communication. By leveraging neuro-symbolic technology integration, the ORKG transforms scholarly articles into a structured, interconnected web of research findings, making scientific knowledge more accessible, discoverable, and actionable. As such, the ORKG is an infrastructure that aims to support the production, curation, publication, and use of FAIR scientific knowledge with a mission to shape future scholarly publishing and communication where the contents of scholarly articles are FAIR research data [2].

The inception of the ORKG is rooted in recognizing the vast, untapped potential of digital scholarship. As researchers around the globe generate vast quantities of data and insights, the imperative to harness this wealth of knowledge in a machine-assisted manner becomes increasingly critical. The ORKG represents a paradigm shift, moving beyond the limitations of traditional research artifacts to a dynamic, open knowledge network. This network not only facilitates the seamless integration, comparison, reproducibility, and machine-based reuse of research findings, but also fosters new collaborations, innovations, and a deeper understanding of complex scientific questions.

Neuro-symbolic integration in the ORKG is a pivotal advancement in achieving these goals because it combines the strengths of symbolic knowledge representation and reasoning with neural networks, creating a synergistic system that enhances the capabilities of the ORKG. The hybrid system leverages the interpretability and rule-based nature of symbolic AI to manage and reason over the structured knowledge graph, while utilizing the pattern recognition and learning capabilities of neural networks to handle unstructured data and improve the extraction and classification of information from scientific articles.

In practice, neuro-symbolic integration in the ORKG involves several key components:

1. *Information Production:* Scientific knowledge is produced symbolically from the outset by integrating symbolic techniques in phases of the research lifecycle, in particular data analysis and manuscript authoring. This reduces the burying of scientific knowledge into narrative text documents [3] and, thus, reliance on information extraction.
2. *Information Extraction:* Neural networks are employed to extract relevant information from scientific articles. These networks are trained on large datasets to recognize and classify various entities and relationships within the text, such as research topics, methods, results, and conclusions.
3. *Expert Curation and Semantic Enrichment:* Once created, the symbolic information is further semantically enriched using symbolic AI techniques and curation by experts. This involves mapping the extracted information to a predefined ontology or schema within the ORKG, ensuring that the information is meaningfully structured and interconnected.

4. *Querying, Reasoning, and Inference:* Symbolic querying and reasoning are used to perform complex queries and infer new knowledge from the existing data in the ORKG. This allows researchers to discover hidden connections, generate hypotheses, and derive insights that would be difficult to achieve through manual analysis alone.
5. *ORKG ASK:* The ORKG Assistant for Scientific Knowledge (ASK, ask.orkg.org) is an advanced interface that utilizes neuro-symbolic integration to provide researchers with intuitive and powerful tools for interacting with the ORKG. It supports natural language queries, allowing users to ask complex questions and receive answers derived from a vast corpus of scientific literature and the integrated knowledge graph. This enhances the usability and accessibility of the ORKG, making it easier for researchers to find and use relevant information.

The integration of these neuro-symbolic technologies transforms the ORKG into a dynamic and intelligent platform, capable of continuously evolving and adapting to the growing body of scientific knowledge. This not only enhances the immediate utility of the ORKG for current research but also lays the foundation for future advancements in scholarly communication and knowledge management. Through neuro-symbolic integration, the ORKG is positioned at the forefront of the transition toward a more interconnected, accessible, and actionable scientific knowledge ecosystem.

Scientific Knowledge Organization (SKO) has undergone significant advancements in recent years, driven by the exponential growth of scientific literature and the need for more efficient methods of managing and accessing this vast body of knowledge. The integration of machine learning and artificial intelligence techniques has started to impact SKO. Recent work demonstrates the effectiveness of deep learning models in automatically classifying and organizing scientific knowledge across multiple disciplines [4]. Approaches using transformer-based architectures have shown promising results in accurately categorizing papers based on content, methodology, and research area [5]. Ontologies continue to play a crucial role in SKO [6] and have reached fields as diverse as behavioral sciences [7], chemistry [8] or bio-diversity and agiculture [9]. Recognizing the collective expertise of the scientific community, there has been a shift towards more collaborative approaches or even crowdsourcing to SKO [10]. Recent efforts have also focused on integrating SKO systems more seamlessly into researchers' workflows, e.g. via workflow management aud reuse [11], holistic data management for semantic interoperability e.g. in humanities [12] or life-sciences [13]. Advancements in data visualization have led to new ways of representing and navigating scientific knowledge [14]. SKO is also impacted by the evolution from basic keyword-based search (such as Google Scholar) to more advanced AI-driven research assistance, progressing through improved metadata utilization (e.g. Semantic Scholar [15]), semantic understanding (e.g. Elicit [16]), and context-aware recommendations. This development pathway represents a beginning shift from simple document retrieval to intelligent, personalized research support systems that understand and anticipate researchers' needs as pursued by the ORKG.

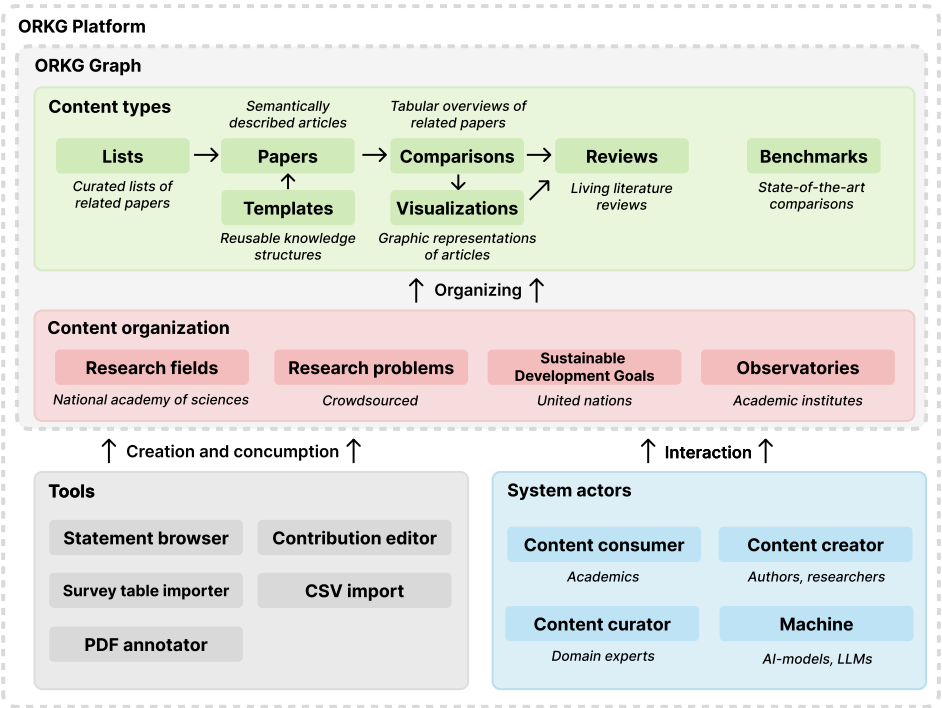

Figure 1. Diagram showing the main ORKG content types. Arrows between content types indicate a workflow for creating literature reviews with the ORKG. Content organization types are used to organize individual content types. Tools are part of the ORKG platform and support data creation and curation. System actors display different actors that are part of ORKG workflows.

2. ORKG Overview

We provide an overview of the core concepts and functionalities of the ORKG, including Benchmarks, Comparisons, Reviews, Observatories, Lists, Content Curation, and Data Use with tools such as Jupyter [17].

Content curation is a critical aspect of the ORKG, ensuring the knowledge graph remains accurate, relevant, and up-to-date. The ORKG supports both manual and automated curation processes. Researchers can contribute by adding, editing, and validating information, while automated tools help extract and update data from various sources. Effective content curation enhances the quality and reliability of the ORKG.

Semantic research contribution descriptions play a crucial role in structuring and standardizing the way research findings are represented. By using predefined templates, the ORKG ensures key elements of research contributions, such as hypotheses, methodologies, results, and conclusions are captured in a consistent and machine-readable format. These templates facilitate the research problem and field-specific annotation of research articles, enabling precise extraction and integration of data. This semantic structuring enhances the discoverability and comparability of research contributions, but also supports automated reasoning and advanced queries.

Lists are curated collections of related research items, such as articles, datasets, benchmarks, or software tools. They serve as an organizational tool to group and catego-

rize information based on specific criteria or themes. Lists can be created and maintained by researchers or automatically generated based on predefined rules and algorithms.

Comparisons are a fundamental feature of the ORKG, allowing researchers to systematically compare research findings in a machine-assisted manner. The ORKG supports the creation of comparative tables and visualizations that highlight similarities and differences between studies. This capability is essential for synthesizing knowledge, identifying trends, and making informed decisions based on comprehensive data.

Reviews facilitate aggregating and structuring information from multiple sources. Researchers can use the ORKG to support systematic reviews and meta-analyses, providing a comprehensive overview of the current state of knowledge on a specific topic. The structured format of the ORKG ensures that reviews are detailed, transparent, and easily editable as new research emerges.

Benchmarks refer to standardized datasets and metrics used to evaluate and compare the performance of various methods and algorithms. The ORKG enables researchers to publish, share, discover, and reuse benchmarks, facilitating the validation and comparison of scientific results across different studies. By providing a common framework for benchmarking, the ORKG promotes transparency and reproducibility in scientific research.

Observatories are specialized knowledge hubs focused on specific research areas or themes. They are domain-specific entry points to the ORKG, providing a centralized platform for monitoring and analyzing trends, advancements, and gaps in research. Observatories are valuable for researchers, policymakers, and other stakeholders seeking to stay informed about developments in specific fields.

Data Use with Jupyter provides robust support for data use and analysis. Researchers can access ORKG data programmatically, enabling them to perform sophisticated analyses, create advanced visualizations, and develop interactive dashboards. Researchers can leverage the extensive ecosystem of Python, R, or comparable packages to enhance their data analysis workflows. The integration with the ORKG allows for seamless data retrieval, manipulation, and visualization, thus supporting the productive use of the ORKG in research.

These functionalities are enabled by the ORKG's robust architecture and modern technologies, which are designed to facilitate efficient knowledge management, data integration, and advanced analysis. A high-level overview of ORKG-related concepts is displayed in Figure 1. The previously discussed content types are part of the ORKG graph itself. In addition, there are content organization types, including research fields, research problems, Sustainable Development Goals (SDGs), and observatories. The content organization entities can be assigned to content types, as a means to organize the content types. Consequently, content types are searchable by their respective fields, addresses, problems, SDGs, and the observatory they belong to. The listed tools in the figure provide user interfaces to consume and curate the ORKG content. Finally, the system actors show the stakeholders of the system and their user interaction roles.

2.1. Architecture and Technology

The ORKG has been developed on top of a modern and scalable architecture while taking advantage of both mature and state-of-the-art technologies integrated in a microservice architecture. The ORKG is based on a knowledge graph – a powerful data model to

structure, interconnect, and semantically enrich information. The way scientific knowledge is modeled in the knowledge graph is similar to RDF [18], in that it is based around the notion of a statement, consisting of two nodes (entities) linked by a directed edge (see Section 3.1). Similar to, but different from RDF, the ORKG uses a Labeled Property Graph (LPG) that is natively more suitable for identifying statements uniquely, and for qualifying these instances through annotating edges/statements. Provenance information is a particular application of such annotations and one of the metadata fields that has been identified as necessary to be connected with statements.

The ORKG ecosystem is designed along two concepts: first, it consists of a microservice architecture, and, second, it follows a classical layered architecture for all its components. This approach enables components and services to scale horizontally while keeping concerns separate. Also, it supports the rapid prototyping and deployment of new features by a larger development/research community, without affecting the performance of the ORKG as a service. Figure 2 gives an overview of the architecture of the ORKG and shows the main components of the platform and their dependencies. This figure also shows that we can add new features driven by real-world use cases while working within the existing functions and services. Furthermore, the microservices paradigm enables the decoupling of services so that they can be developed, deployed, and scaled independently of each other. A modular architecture ensures flexibility and maintainability, so that problems in one service do not affect the entire system.

In addition, each ORKG component follows a layered architecture (data access, business logic, and presentation), thus adhering to the separation of concerns. This architectural pattern helps developers achieve code organization and testability, and enables the reuse of different components or services. By following these software engineering principles, the ORKG system can evolve with changing requirements, adapt to new technologies, and scale to cope with increasing workloads and data volumes. The use of a microservice architecture combined with the layered architecture for each component provides a robust, extensible, and sustainable system capable of meeting the requirements of the research community.

In the realization of this architecture, the ORKG uses a very heterogeneous technology stack. At the core, the backend is implemented in Java, with the Spring Boot framework, guaranteeing good support for REST APIs and component communication, along with the smooth integration of external services. Additional REST APIs are exposed in Python, which is used for functionality that either requires higher flexibility or needs specialized libraries. At the center of the ORKG is a Neo4J-based[2] knowledge graph that enables both effective storage and querying for complex scholarly relationships. The Qdrant[3] vector store enriches the graph structure with advanced search functionality, such as similarity search and semantic matching. Furthermore, the frontend is powered by modern web technologies that guarantee good user experience across devices: ReactJS[4] for dynamic user interfaces, NextJS[5] for server-side rendering, and Tailwind CSS[6] for responsive design and styling web pages.

[2] `https://neo4j.com/`
[3] `https://qdrant.tech/`
[4] `https://react.dev/`
[5] `https://nextjs.org/`
[6] `https://tailwindcss.com/`

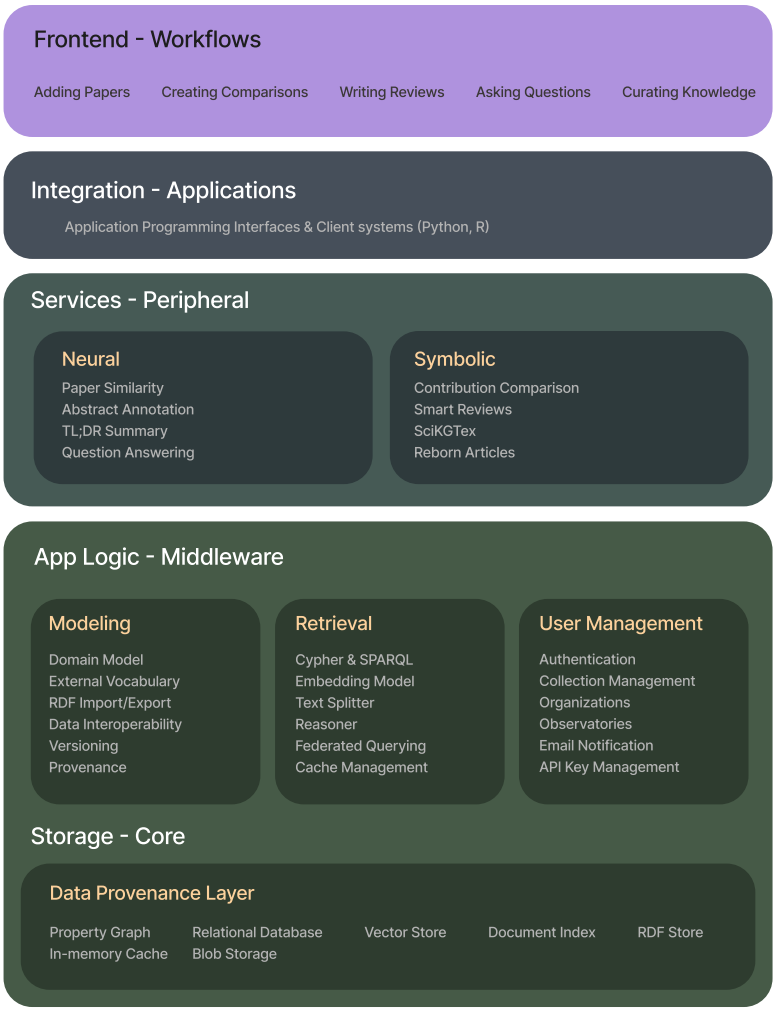

Figure 2. The ORKG technological architecture. The diagram highlights the main components of the ORKG ecosystem and the layered dependencies between the various components.

The ORKG promotes open science and transparency; hence, the codebase is completely open source. Such commitment furthers collaboration, innovation, and gives researchers and developers around the world the opportunity to audit, improve, and further adapt the platform. With open source development, the ORKG stands for a community-driven platform that evolves to meet the demands of the research community.

3. Symbolic Sub-system

The symbolic sub-system of the ORKG comprises the generic knowledge model, a templating system for specifying domain-specific knowledge, and a method for symbolic knowledge production tightly integrated into the phases of the research life-cycle.

3.1. Symbolic Knowledge Model

We briefly describe the main types that constitute ORKG's symbolic knowledge model, specifically statements, research contributions, papers, and comparisons.

Inspired by the RDF data model, the *Statement* is the basic type of the symbolic knowledge model in the ORKG. A Statement s_t is a triple $s_t = (s, p, o)$ for the statement's subject s and object o related by a predicate p. In terms of graph data model, s, o are KG nodes and p is a directed KG edge relating s with o. Logically, subjects s are ORKG Resources (i.e., instances, named individuals), predicates p are ORKG Predicates (aka Properties), and objects o may be ORKG Resources or ORKG Literals.

Building on statements, the ORKG symbolic knowledge model defines the *Research Contribution* r_c as the tuple $r_c = (\mathbf{S_t}, \mathbf{R_m})$, whereby $\mathbf{S_t}$ is a set of statements $s_t \in \mathbf{S_t}$ and $\mathbf{R_m}$ is a set of metadata associated with the research contribution r_c. ORKG does not further specify the set $\mathbf{S_t}$ of a research contribution. The general guideline is for a set $\mathbf{S_t}$ to describe a research finding originally published in an article, including the addressed research problem and the utilized materials and methods. Users may specify sets $\mathbf{S_t}$ using ORKG Templates (see Section 3.2). In the ORKG, the metadata set $\mathbf{R_m}$ consists of classic bibliographic metadata elements such as title, authors, and identifiers as well as provenance metadata elements including creation time and contributing user.

Building on research contributions, the ORKG symbolic knowledge model defines two primary derivative types: the ORKG *Paper* p_r as the tuple $p_r = (\mathbf{R_c}, \mathbf{R_m})$ and the ORKG *Comparison* c_r as the tuple $c_r = (\mathbf{R_c}, \mathbf{R_m})$, whereby $\mathbf{R_c}$ is a set of research contributions $r_c \in \mathbf{R_c}$ and $\mathbf{R_m}$ is a set of metadata associated with the paper p_r or the comparison c_r, respectively. ORKG papers and comparisons are thus both aggregations of research contributions. The main difference is that for papers p_r the aggregation consists of research contributions as originally published in a *single* research article while for comparisons c_r the aggregation consists of research contributions that share a common dimension (typically the research problem), but are originally published in *multiple* research articles[7]. ORKG papers are thus descriptions of the research findings originally published in the corresponding articles while ORKG comparisons juxtapose research findings to enable overviews of the state-of-the-art.

Describing the research findings originally published in an article in the ORKG involves the task of semantic parsing, i.e., determining how to model the semantic content of specific natural language statements into a graph representation by choosing appropriate ontology terms and graph patterns (i.e., semantic data schemata). Such graph patterns can be specified using constraint languages such as the Shapes Constraint Language (SHACL; see also Section 3.2).

Due to the high expressivity of the underlying triple structure (s, p, o) of the graph, any given natural language statement can be modelled in numerous and fundamentally

[7] As a corner case, it is possible to compare two or more research contributions originally published in a single research article, but in practice this is uncommon.

different ways. If a knowledge graph allowed more than one semantic pattern for modelling the same type of information and if it did not restrict the choice of ontology terms to be used in such a pattern, it would encounter substantial semantic interoperability issues regarding terminological and propositional interoperability, resulting in lack of overall findability, interoperability, and reusability of information within the graph [19].

Knowledge graphs typically attempt to model a mind-independent reality, with nodes representing real-world entities or concepts and the edges representing the relationships or associations between them. Many knowledge graphs apply this modelling paradigm and restrict their content to a specific scope, thereby typically employing a *static information extraction approach* in which they model a fixed set of statement types using a fixed set of associated graph patterns and ontologies. This *static knowledge graph construction approach* allows for the pre-definition of all necessary graph patterns and ontology terms, ensuring a logically consistent and semantically interoperable structure, resulting in a truly FAIR knowledge graph that supports reasoning. For ontology engineers closely collaborating with respective domain experts it is feasible to specify the limited set of graph patterns required for modelling all relevant information that is in the scope of the knowledge graph and to agree on the ontology terms being used with these patterns.

With its open, cross-domain scope, the ORKG, however, cannot adhere to the conventional static information extraction approach. Instead, it follows a *community-driven dynamic knowledge graph construction (DKGC) approach*, where the graph's information and knowledge are continuously evolving. DKGC approaches present unique challenges, as it is not feasible to pre-define all graph patterns and ontology terms required for modelling the information and knowledge that ORKG users might want to add. Consequently, the task of semantic parsing is imposed on the user, usually without support from an ontology engineer. This shift represents a significant barrier to data entry into the ORKG and is likely to lead into semantic ambiguities, logical inconsistencies across the knowledge graph, general data quality issues, and a lack of semantic interoperability and FAIRness of its data, resulting in limited findability of knowledge within the graph. Additionally, DKGC approaches usually require versioning of the graph's contents, as users may modify them at any time. Ideally, this versioning would include a detailed editing history to provide transparency and build trust among users.

Another consequence of applying the DKGC approach in cross-domain knowledge graphs is that reasoning over the entire graph is out of scope for both practical and theoretical reasons. Practically, it is not feasible to develop graph patterns across all domains that are logically consistent using ontology terms. Theoretically, constructing a set of ontologies that are both logically consistent and meet domain-specific requirements is impossible. For instance, Newtonian physics and quantum physics are ontologically inconsistent, yet physicists rely on both. However, when achieving reasoning capability is no longer a viable goal, tuples need not necessarily reflect the underlying semantics of a mind-independent reality. This opens new possibilities for modelling the information and knowledge in the graph.

To address the challenges of applying the DKGC approach, we developed and are currently implementing the *Rosetta Statement (RS) Approach* [20] for modelling natural language statements. The core idea is that statements are the smallest units of information that are semantically meaningful to a human. By adopting the RDF reification approach and applying it to simple natural language statements instead of triples, we de-

fined a very general model that can be applied to any natural language statement comprising a subject phrase, a predicate/verb, and one or more object phrases. Each statement is represented in the graph by its own ORKG Resource, to which respective subject and object resources—instantiating appropriate ontology classes—are linked, enabling straightforward modelling of n-ary relations. The statement resource instantiates a corresponding RS class, defined based on the underlying predicate/verb. To each RS class, a graph pattern in the form of an ORKG template is associated. Since all RSs follow the same general underlying graph pattern, users can specify new types of RS classes and accompanying graph patterns by answering a set of simple questions, requiring no background in formal semantics and no support by ontology engineers. For each Rosetta template, we also specify a dynamic label that provides a human-readable sentence, with the subject and objects being variables identified by parsing the labels of the corresponding subject and object resources and literals of a given RS. Consequently, all content in the ORKG that has been modelled using RSs will be human-readable. Moreover, the approach supports versioning and provides a detailed editing history for each RS.

Modelling natural language statements instead of a mind-independent reality offers several advantages. It simplifies the semantic parsing task, lowering the barrier for adding content to the ORKG. It also provides a universal and generally applicable graph data structure against which functions can be programmed without having to consider the various peculiarities of domain-specific graph patterns, as it would be the case when modelling a mind-independent reality. Furthermore, the structural proximity of RSs to natural language statements facilitates the involvement of Large Language Models (LLMs) in developing services and functions to support users in any task that involves interacting with the graph. And although graphs based on the Rosetta approach are not capable of reasoning, reasoning capability can still be achieved for subgraphs within the ORKG by defining schema crosswalks between selected RS templates and graph patterns that support reasoning.

3.2. Specifying Domain-specific Symbolic Knowledge with Templates

Building on the Shapes Constraint Language (SHACL) [21] to define and enforce data structures, ORKG templates enable the definition of domain-specific knowledge types and support data consistency and quality.

While the ORKG does not implement SHACL in its entirety, it utilizes SHACL's vocabulary to guide the setup of input forms and validate data. This approach maintains data quality and structure without the need for full SHACL implementation on the graph and by adopting SHACL, the ORKG ensures that data adheres to specific formats and constraints, facilitating interoperability and enhancing the reliability of scientific knowledge integration. This approach allows machines to effectively understand and compare knowledge originally published in different articles.

The overall template system function is illustrated in Figure 3. Domain experts create templates that define the knowledge type of research contributions, integrating these templates with ontologies through the TIB Terminology Service[8]. This integration enriches the templates with standardized vocabularies, ensuring that the data captured is consistent with broader semantic frameworks.

[8] `https://terminology.tib.eu/ts`

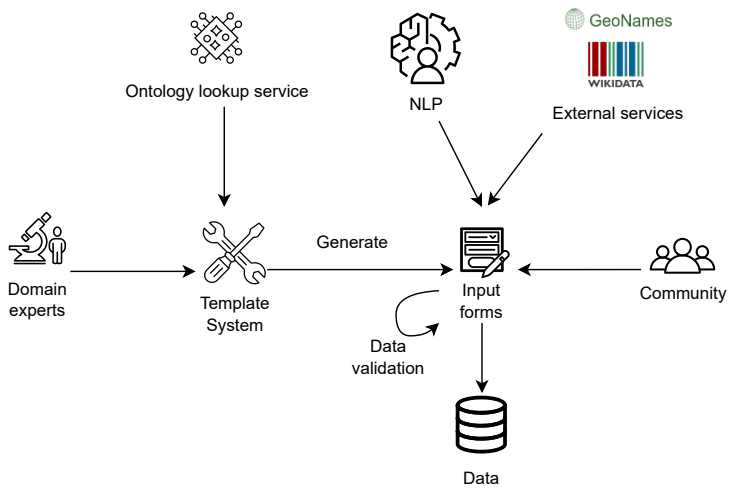

Figure 3. Schematic overview of the ORKG template system.

Once a template is created, the ORKG's Statement Browser component (Section 3.3) parses it, setting up the appropriate properties and constraints to generate specialized input forms. These forms ensure that data entered by researchers adheres to the template's specifications. Additionally, the system assists users by providing suggestions from the ORKG NLP service and offering autocomplete functionality that suggests values from well-established knowledge graphs such as Wikidata[9] or Geonames[10] for location values.

Before storing any new data, the Statement Browser validates it against the template's constraints, ensuring that it meets the expected data types, formats, and relationships. This validation process is crucial for maintaining the integrity of the data within the ORKG. Furthermore, it assists users by providing suggestions from the large language model (Section 5.2), ensuring that the introduced values accord with guidelines[11], especially when no restrictions have been set by the domain expert while creating the template.

The ORKG template system incorporates the ORKG NLP system, which assists users in selecting the most appropriate template for their research. When a researcher adds a paper, the system suggests a template based on the paper's research field, title, and abstract. This feature streamlines the data entry process and ensures that the data is organized in a manner that aligns with the article's content.

The template visualization diagram is an integral part of the template editor. This feature offers a UML-like representation to visually map the structure and relationships of templates. The diagram aids in comprehending complex template designs, allowing users to grasp the overall schema and connections more intuitively. Figure 4 presents the different components of the diagram. Each entity in the diagram represents a template in the ORKG. The header displays the template name and its associated target class, while the property relationships are presented visually through edges linking the properties to their respective template entities. The template visualization diagram is a powerful

[9] https://www.wikidata.org
[10] https://www.geonames.org/
[11] https://orkg.org/help-center/article/42/Modeling_best_practices_for_resources_and_properties

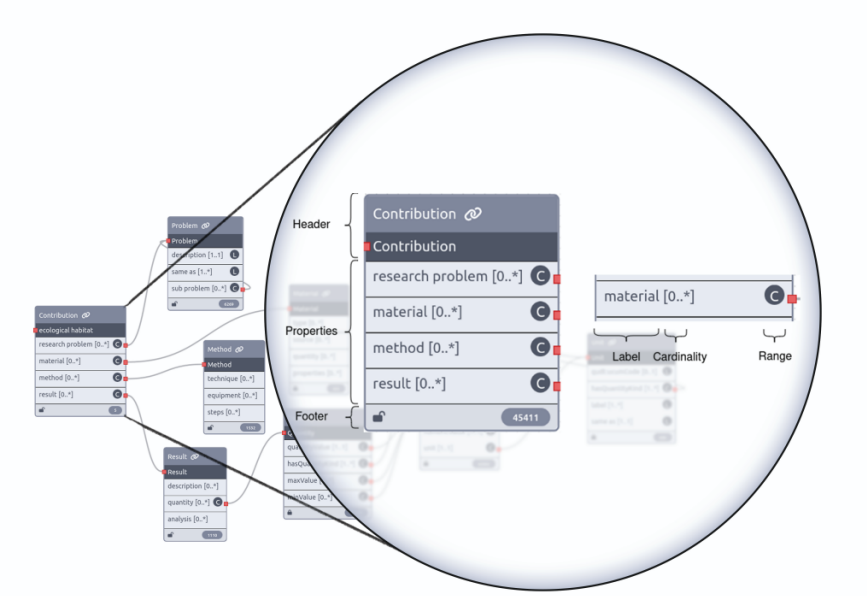

Figure 4. ORKG template diagram with a zoom in on one of the entities.

tool for quickly understanding the structure of templates, making it more accessible and efficient for users to analyze, create, and modify templates.

3.3. Symbolic Knowledge Production During Phases of the Research Life-cycle

Since scientific knowledge is generally available in unstructured form, producing structured symbolic knowledge is a key challenge for the ORKG. We have pursued related techniques falling into two broad categories: (1) expert-driven structuring, including author-driven at the time of knowledge production (i.e. pre-publication during phases of the research life-cycle) and (2) machine-driven structuring via knowledge extraction from articles (post-publication). Knowledge extraction is powered by ORKG's neural sub-system presented in Section 4. In this section, we present the expert-driven category of techniques, namely the statement browser in ORKG's frontend, SciKGTeX supporting annotations in LaTeXmanuscripts, and reborn articles enabling symbolic knowledge production in data analysis and final data production.

Statement Browser. The ORKG statement browser is a component of the Web-based frontend supporting the expert-driven manual description of research contributions as symbolic scientific knowledge. Knowledge is described as sets of statements, each consisting of a subject, property, and object. The frontend displays the subject in the header of the statement browser, and the property and object as columns. This simplifies user input, as data input is similar to providing key-value pairs, for the property and object positions, respectively. To further simplify user input, the statement browser integrates the ORKG template system presented in Section 3.2 to assist user input with user-defined knowledge types that specify properties and their constraints, presented by the frontend to users as forms. Furthermore, the statement browser integrates different data viewers, depending on the data type. This includes the display of data tables, images, videos, expressions in LaTeX, maps, etc. To reflect state-of-the-art user interaction patterns, prop-

erties and resources are selected via an autocomplete input, guiding users into reusing existing terms when available. To enable suggestions and improve interoperability, the autocomplete input integrates with external terminologies, in particular Wikidata, GeoNames, and ontologies from the EBI Ontology Lookup Service (OLS). It is thus possible to discover and select required terms without leaving the ORKG frontend. As an additional feature, the data type selector supports specifying the data type of objects. This includes resources (via the autocomplete) and literals (i.e., text, decimal, boolean, date, etc.). Additionally, special data types such as `Empty` and `List` allow for indicating that certain data are not reported in an article or support creating ordered lists. Finally, the statement browser also includes Smart Suggestions, as discussed in Section 5.1. These suggestions provide guidance to users while interacting with the statement browser. The statement browser is a reusable component, integrated at various locations in the ORKG frontend.

SciKGTeX. Similar to systems such as dokieli [22], SciKGTeX [23] supports symbolic knowledge production at the time of manuscript authoring during the scholarly communication phase of the research lifecycle. It is a LaTeXpackage for annotating research contributions directly in LaTeXsource code [24], based on the idea of "FAIR-by-Design" artifacts [25]. Authors can describe their manuscript with annotations and thereby create symbolic scientific knowledge that describes their research contributions at the time of writing. SciKGTeX embeds the annotations into the PDF's XMP metadata so that the symbolic knowledge persists with the manuscript. From the perspective of the research lifecycle, SciKGTeX anchors symbolic knowledge production in the scholarly communication phase and supports authors in creating "FAIR-by-Design" manuscripts [25]. These "FAIR-by-Design" manuscripts simplify the storage of the symbolic knowledge produced, which leads to better access to the knowledge of the manuscripts and thus lays the foundation for their effective (re-)use. The ORKG offers an upload function for SciKGTeX-annotated PDFs so that authors can easily add symbolic knowledge to the ORKG. Thus, the ORKG demonstrates how infrastructure providers can leverage such "FAIR-by-Design" manuscripts to automatically ingest symbolic knowledge in their systems and support dissemination, discoverability, and (re-)use of symbolic scientific knowledge.

Reborn Articles. The mainstream approach to scientific knowledge graph construction, i.e. symbolic scientific knowledge production, relies on knowledge extraction from published articles, either manually or supported by machines using NLP/LLM neural techniques (see Section 4). By enabling the structuring of knowledge at the time of knowledge production, specifically final data production in data analysis, reborn (born-reusable) articles [26] also fall into the category of expert-driven techniques that do not rely on knowledge extraction from articles. The primary intuition is that ORKG templates and, more generally, user-defined and well-specified data types accessible via Data Type Registries [27] including Schema.org, can be leveraged in any phase of the research lifecycle to ensure the efficient production of accurate and rich symbolic scientific knowledge. In addition to data analysis and manuscript authoring, this notion of "FAIR-by-Design" is applicable also earlier in the research lifecycle, e.g. in experiment and data management planning or in electronic lab notebooks that document experiment execution.

4. Neural Sub-system

Previous sections explored human curation methods for representing symbolic knowledge in the ORKG. This section transitions to discussing the automation goals against the backdrop of the ORKG, specifically those achieved using neural-network-based systems.

4.1. Generic Knowledge Extraction

As a start to the discussion on automation goals, this section highlights the importance of automated research field classification for scientific articles in the digital libraries generally as well as for the ORKG that are achieved using neural-network-based methods. Research field classification, which involves categorizing academic publications, represents a highly structured and specialized application within the broader context of generic knowledge extraction [4]. While generic knowledge extraction covers a wide range of techniques and applications across various data types and domains, research field classification focuses specifically on the academic domain, using similar techniques to achieve specialized goals in organizing and understanding scientific literature. As a subdomain of generic knowledge extraction, research field classification extracts textual features indicative of specific research fields by identifying relevant keywords, phrases, and contexts characteristic of certain disciplines [28,29,30,31]. This task enhances academic search engines and digital libraries [32,33,34,35,36] by enabling precise and relevant searches based on specific research fields. Furthermore, it supports bibliometric analysis, trend identification in research, and interdisciplinary studies by providing detailed classifications of scientific articles [37].

Recent advancements in neural systems for research field classification have significantly improved the precision and efficiency of categorizing scientific articles. One notable study introduces SCINOBO, a novel system that uses neural topic modeling [38] and community detection to dynamically construct a hierarchical Field-of-Science taxonomy, enhancing the classification of scientific papers across various domains [39]. This approach extends the taxonomy up to five levels, incorporating OECD fields [40] and demonstrating robust performance in classifying publications within the knowledge domains of Energy and AI. Additionally, a hierarchical classification system using deep learning, detailed by Rao et al. [41], categorizes research into a three-tier system, achieving high accuracy in distinguishing disciplines, fields, and subfields across 160 million abstracts. This system is designed to support both single-label and multi-label classification settings, showing over 90% accuracy in numerous experiments.

Language models, advanced systems using neural network architectures like transformers [42], are at the forefront of AI research and widely used for research field classification. Trained on vast amounts of text, they are specifically designed to generate and understand natural language, crucial for accurate text processing. These models utilize word embeddings, which are vector representations of words that capture their meanings, syntactic roles, and relationships with other words, enabling deeper semantic understanding. E.g., BERT (Bidirectional Encoder Representations from Transformers) [43] and its scientific variants [44,45,46], or the GPT series (Generative Pre-trained Transformer) [47,48,49]. Language models possess several key features in support of research field classification: 1) contextual understanding, which allows them to distinguish between different fields using similar terminology based on surrounding text; 2) semantic similar-

ity, enabling accurate grouping of papers by understanding underlying topics and themes despite variations in vocabulary; 3) adaptability to new terminology, ensuring classifications remain current with evolving language use by training on diverse and recent datasets; 4) handling ambiguity and polysemy by using contextual clues to disambiguate terms with multiple meanings; 5) interdisciplinary insights, facilitating the classification of cross-disciplinary research by recognizing connections between different fields; and 6) scalability and efficiency, allowing rapid and comprehensive analysis of large volumes of research papers, crucial for managing the exponential growth of scientific literature. By leveraging these capabilities, language models can significantly enhance the accuracy and reliability of research field classification, supporting better organization and retrieval of scientific knowledge.

Building on these capabilities, the study by Wolff et al. [50] illustrates a practical application of language models for research field classification against the ORKG fields taxonomy. They employ a method using enriched BERT embeddings to classify articles into one of 123 predefined and representative fields from the ORKG taxonomy of over 700 research fields. Utilizing English scientific articles sourced from ORKG and arXiv, the team enhanced the dataset with metadata from bibliographic databases such as OpenAlex, Semantic Scholar, and Crossref. By leveraging various pre-trained language models, particularly BERT and its scientific variants like SciBERT, SciNCL, and SPECTER2 [51], the study focused on fine-tuned transfer learning approaches. The inclusion of additional metadata significantly improved classification accuracy, with the SPECTER2 model achieving the highest performance with a weighted F1-score of 0.7415. This approach demonstrates how combining pre-trained models with strategic data enrichment can further harness the inherent capabilities of language models, enhancing the precision and utility of automated scientific article classification systems.

4.2. Domain-specific Knowledge Extraction

Essential to support the automated construction of KGs such as the ORKG are also research-domain-specific automated KG extraction methods.

The emergence of the "deep learning tsunami" [52] revolutionized KG construction pipelines, which led to the adoption of neural learning techniques. Traditionally, this resulted in hybrid architectures that blended neural and symbolic methods, utilizing a sequence of complementary deep learning and rule-based strategies across different stages of the knowledge composition process to finally arrive at a KG. The CS-KG system [53] or AI-KG system [54] serves as a notable example of neural-symbolic scientific KG construction. It incorporated the DyGIEpp deep learning module [55], which operated within the transformers architecture [43] to extract predefined entities and relations. Additionally, it employed the Computer Science Ontology classifier (CSO-C) [56] for enhanced entity extraction. The system utilized the Stanford Core NLP suite's OpenIE [57] to identify open domain relations, and the Stanford POS Tagger [58] to pinpoint verbs between entity pairs as potential relation candidates.

In general using the pipelined approach For KG construction, standard components include named entity recognition (NER) and relation extraction (RE) tasks, which can be efficiently handled by fine-tuning transformer models based on the BERT architecture [43] on specific datasets. BERT utilizes pretrained parameters from large-scale corpora like Wikipedia or Books, trained with a masked language model objective, en-

abling robust natural language understanding. The fine-tuning procedure involves initializing the desired language model with these pretrained parameters and then tuning the probabilistic parameters within task-specific architectures for downstream extraction tasks such as NER or RE using task-specific datasets. In the scientific domain, widely used transformer models include SciBERT [44], PubMedBERT [59], SemMedDB [60], BioBERT [61], BioClinicalBERT [62], and BlueBERT [63].

Additionally, in scientific KG construction, it is common to leverage external ontologies and knowledge bases for entity and relation extraction and linking, a practice often observed in biomedicine. Ontologies facilitate standardized representations and ensure interoperability between KGs, thereby enhancing their effectiveness and utility in complex research settings. For instance, the iASiS KG [64], which extracts knowledge from biomedical scientific articles on lung cancer and dementia, utilizes NLP techniques and standardized biomedical ontologies such as UMLS [65] to annotate entities and relations. This particular KG focuses on capturing knowledge about drug interactions in treatments for lung cancer, which also defines its scope for knowledge extraction. Another example is OpenBioDiv [66] in the domain of biodiversity science, where the knowledge extraction objective is guided by the OpenBiodiv-O ontology [67]. Consequently, the focus of knowledge extraction for KG construction varies by domain, ranging from diseases in biomedicine to plant treatments in biodiversity.

As introduced earlier, the ORKG aims to capture the contributions or structured salient facets of research. In this vein, showcased in the next sections 4.2.1 and 4.2.2 are two different domain-specific knowledge extraction objectives that have been explored for the ORKG.

4.2.1. Leaderboards Extraction

In the field or domain of artificial intelligence (AI), researchers commonly develop new models aimed at achieving state-of-the-art (SOTA) performance, reported through four key elements: Task, Dataset, Metric, and Score (T, D, M, S). Recently, it has become standard to compile these details into community-managed leaderboards, which function like scoreboards, showcasing the scores of various AI models for specific tasks, datasets, and metrics. The ORKG platform exemplifies this with its benchmarks feature. Figure 5 highlights the shift from traditional article-based reporting of (T, D, M, S) to recording them in the ORKG leaderboards. Moving away from the traditional community-curated approach, automated text mining techniques are now prevalent. The goal of such automation is to develop systems that can analyze the full text of AI research papers, determine if they report model performance on benchmark datasets, and extract all relevant (T, D, M, S) tuples [68,69].

Language model based neural systems have been shown to be efficacious for the (T, D, M, S) knowledge extraction task. The focus of these automated tasks has shifted from entailment testing [71,70,72] to generation tasks [73,74,75], driven by advancements in large language models (LLMs). Knowledge extraction systems developed by Kabongo et al. [73,74] are notable advancements. Their recent approach involved leveraging the FLAN-T5 strategy [76], which encompassed fine-tuning a pre-trained LLM with a standard set of instructions to better equip it for various tasks. Using the applicable instructions from the FLAN-T5 collection, they fine-tuned LLMs such as Mistral-7B [77] and LLaMA 2 [78], to improve their suitability for the (T, D, M, S) extraction task. For this fine-tuning, they employed the QLoRA (Quantized Low-Rank Adaptation) technique

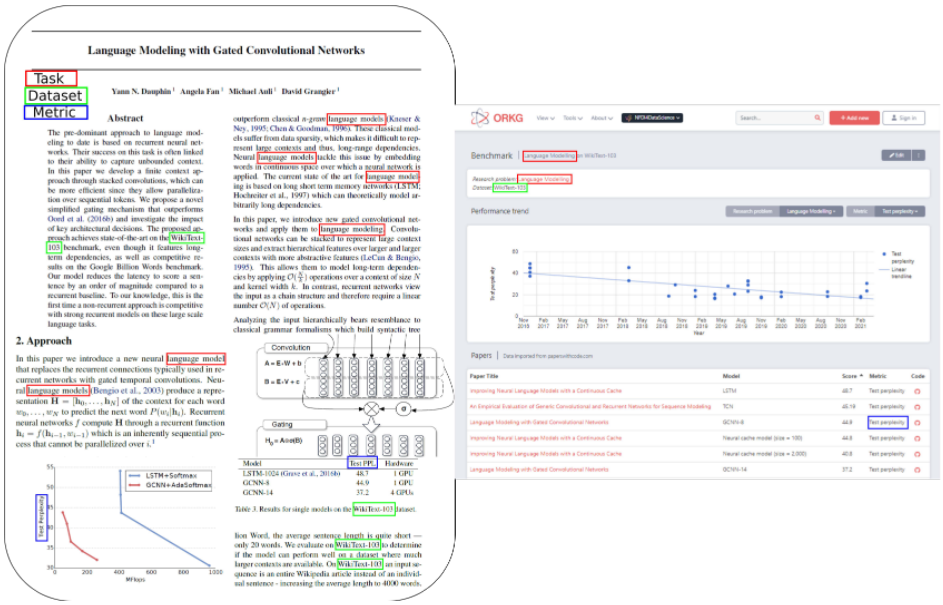

Figure 5. This figure presents a side-by-side comparison: on the left, it shows task, dataset, and metric information published in the traditional PDF format as non-machine-actionable data; on the right, the same information is displayed as machine-actionable data, recorded within the leaderboard of the ORKG benchmarks feature. This figure is reproduced from the original manuscript [70].

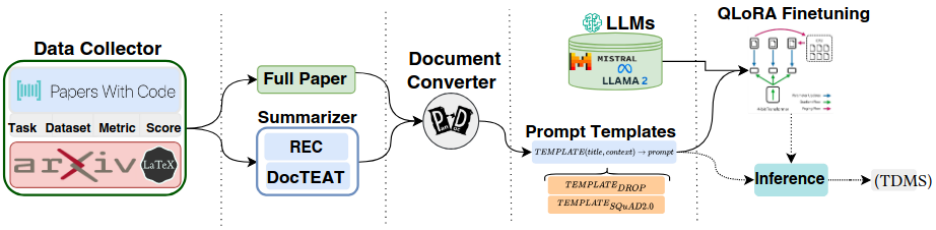

Figure 6. The automated (Task, Dataset, Metric, Score) knowledge extraction task workflow with large language model (LLM)-based neural systems. This figure is reproduced from the original manuscript [73].

[79], which reduces the computational cost and memory footprint by efficiently adapting the model using quantization and low-rank approximation techniques. Figure 6 illustrates the overall workflow of the approach. To identify the optimal context for (T, D, M, S) knowledge extraction, they tested heuristic-based methods focusing on parts of the article most likely to contain this information. They experimented with DocTAET (title, abstract, experimental setup, and tabular information), DocREC (results, experiments, and conclusions), and DocFULL (entire text). Among these, DocTAET proved to be the most effective. For additional experimental insights, we refer the reader to the original paper [73].

4.2.2. Basic Reproduction Number Estimate (R0) Extraction

As a second showcase, we address domain-specific knowledge extraction for the ORKG, focusing on a key research problem in Epidemics and Virology: the basic reproduction number estimate ($R0$) for infectious diseases. The $R0$ estimate represents the average number of secondary infections caused by a single infected individual [80] and was a key informat used by the responsible government organizations during the Covid 19 pandemic. The extraction objective was to develop an automated system to extract the $R0$ estimate values including its other relevant properties to add to the ORKG. Following the (T, D, M, S) extraction objective discussed earlier, and based on a preliminary research comparison[81], the comprehensive knowledge extraction target included six properties: disease name, location of study, date of study, $R0$ estimate value, %CI (confidence interval) values, and the method used to estimate the $R0$.

Despite current advancements in LLMs, they are noted to still struggle with specialised domains like science [82,83]. Efforts to align LLMs with scientific demands for knowledge extraction are in early stages, focusing on domain-specific fine-tuning [84]. To address the $R0$ estimates knowledge extraction task highlighted in this section, Shamsabadi et al. [85] instruction-finetuned an LLM, specifically the 780M parameter Flan-T5 variant [76], to align it with the extraction objective. They subsequently created a queryable knowledge graph (KG) as part of the ORKG for virologists [86]. Some of the representative infectious diseases extracted were covid-19, dengue, influenza, HIV, SARS, MERS-COV, cholera etc. Based on evaluations of the model's extraction performance on a gold-standard corpus of 1,500 annotated abstracts [87], their model (ORKG-FLAN-T5$_{R0}$) outperformed the vanilla variants of the finetuned models, i.e., T5 [88] and FLAN-T5, as well as a state-of-the-art GPT model with over a billion parameters, despite having significantly fewer parameters than GPT. This work, along with similar studies in biomedicine [89] and materials science [90], suggests that domain-specific instruction-finetuning of LLMs supports transfer learning and enhances knowledge extraction performance.

4.3. Neuro-symbolic Question Answering on Millions of Publications with ORKG ASK

The ORKG offers innovative approaches for interacting with scientific knowledge. Among these, the ORKG Assistant for Scientific Knowledge (ASK, ask.orkg.org) stands out as a modern tool for researchers and academics. ASK provides an intuitive natural language interface that enables users to search through millions of indexed and cataloged scientific items with ease and precision.

Embedding-based Scientific Literature Search. Traditional search engines generally only leverage keyword matching whereas the technology behind ASK permits input of full research questions and by leveraging document embeddings in turn greatly improves the relevance and accuracy of search results. This approach minimizes time and effort spent on locating relevant information, helping researchers to focus on their primary work instead of sorting among tons of irrelevant documents. ORKG ASK is not limited to classic search engine constraints, it actively helps to filter through the massive corpus of scientific literature and in practice, produces a set of documents that are directly relevant to a researcher's inquiry. In addition, it automatically generates important information based on these documents to assist researchers in their knowledge discovery.

ASK Overview. ASK is built upon three fundamental components, each playing a crucial role in its functionality:

- *Semantic Search:* Utilizes advanced natural language processing techniques to understand the context and meaning behind user queries.
- *Large Language Model (LLM):* Employs state-of-the-art AI to process and analyze scientific texts.
- *Knowledge Graphs (KGs):* Organizes and interconnects scientific information in a structured, easily navigable format.

The typical ASK service interaction can be summarized as follows:

1. *Query Input:* The researcher starts by formulating a research question.
2. *Semantic Parsing:* A semantic search component parses the question and represents it in a dense numerical vector capturing its essence.
3. *Corpus Comparison:* The resulting vector representation is then compared with a vast corpus of indexed documents, which have been similarly embedded.
4. *Relevant Article Retrieval:* Using Approximate Nearest Neighbor (ANN) algorithms [91], ASK identifies the most relevant articles within the corpus.
5. *Information Extraction:* Once relevant articles are found, a series of tailored prompts guide the generation of important details based on the articles' content.
6. *Retrieval Augmented Generation (RAG):* Built on top of open-source LLMs (e.g., Mistral [77]), the RAG [92] process ensures that the generated information comes only from the retrieved articles, minimizing the hallucinations of content. RAG does not ensure verbatim generation from the text, rather it limits the model to use in-context learning compared to parametric knowledge.
7. *Results Presentation:* Items of interest and extracted information are presented in tables that display item-specific formats, similar to traditional literature review tables.
8. *Knowledge Graph Integration:* The knowledge graph can be used for additional processing to refine results or filter the search space. It also acts as a core, linking the generated content to third-party knowledge graphs through Linked Open Data entities such as DBpedia [93].

Figure 7 shows a high-level depiction of a question answering workflow that invokes multiple components of the system in order to find relevant documents and answer the posed research question with citable claims.

Functionalities Beyond Search. Though various steps and components are involved from the moment a research question is presented to the system and the relevant documents are retrieved and augmented with relevant extracted information, ASK is a user-functionality-rich tool. Figure 8 shows the design of the ASK user interface. The interface is designed to be simple and accessible to all users but still capable of a wide array of functionalities.

We tag the figure with multiple nodes to highlight the actions that users can perform. *Node 1* (Question Answering) displays the input field for the research question. The answers to the question are presented in two ways. *Node 2* offers a synthesized response based on the top five resulting documents from the semantic search, while *Node 3* displays answers (if available) based on the individual papers. *Node 4* highlights the information extraction aspect of the workflow. Beyond answering the research question,

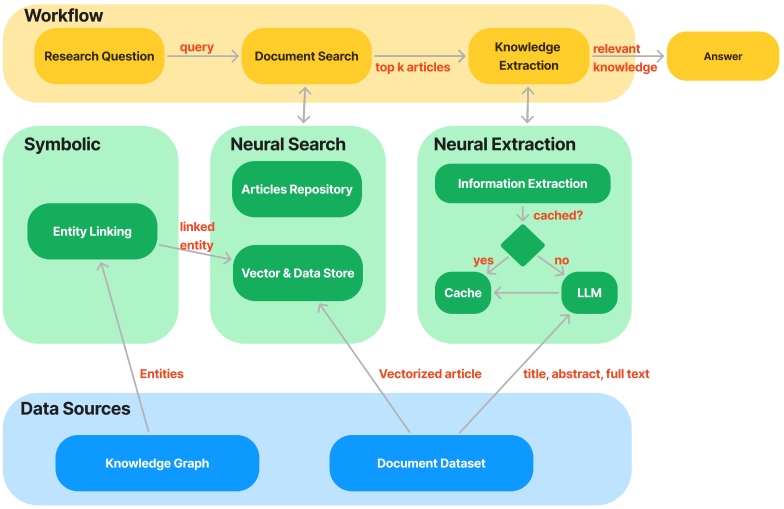

Figure 7. High-level sketch of the ORKG ASK workflow. The workflow shows how the system invokes both symbolic and neural subsystems to answer natural language questions from its internal document index.

additional information is extracted from relevant papers and displayed in tabular format (columns) next to the answer *Node 3*. A researcher is able to use this functionality to extract particular pieces of information that are more relevant to their search (e.g., dataset, participant count, etc.).

Given the large number of research articles out there and the sizable subset indexed within ASK, the use of filters is necessary to further narrow down the search space. *Node 5* exposes a fine-grained filtering system that enables users to refine their search based on various criteria such as language, year, citation count, and author names. This functionality can be used in conjunction with the bibliography manager that is integrated in the user interface *Node 8*. The manager empowers users to store and organize articles in collections under "My Library". As such articles can be added via the bookmark icon in the interface *Node 8*. Furthermore, it is possible for user to incorporate articles into My library by using DOI, title, or BibTeX import. These manually added articles are visually distinguished (by color) in the interface to provide feedback to the user.

To facilitate data reuse, articles are represented in CLS-Json format, providing standardized bibliographic metadata. With *Node 7* users can cite data items in various styles such as APA, Harvard, etc., and export citations in multiple formats including BibTeX and CLS-Json. Additionally, the entire search result table can be exported to CSV and ORKG CSV formats as highlighted with *Node 6*. To further enrich the data, an Entity Linking pipeline continuously applied to abstracts enables filtering articles based on semantically identical concepts that are mentioned in the relevant items.

In this section, we presented the details of ORKG ASK, a neuro-symbolic question answering tool. ASK with the integration of semantic search, large language models, and knowledge graphs, provides a novel solution for researchers to process vast amounts of scientific literature. In addition to saving time, ASK expands the scale of analysis in lit-

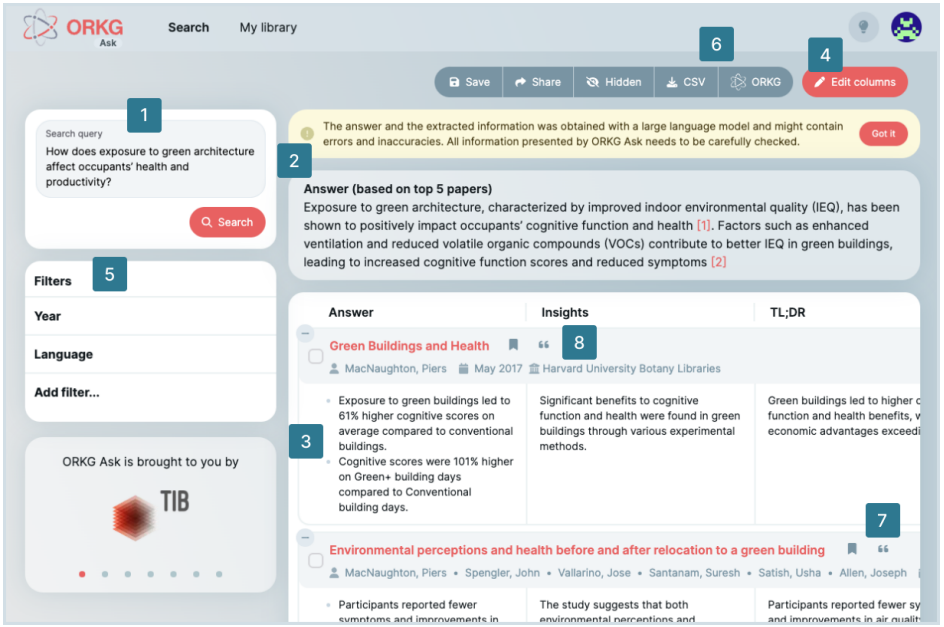

Figure 8. Design of the search result page of the ORKG ASK system. Different interaction opportunities for users are highlighted on the interface in numbered blue nodes.

erature reviews, potentially leading to new findings and connections in various fields of study. With the increasing volume of published scientific articles, tools such as ORKG ASK are necessary to ensure relevant research is easily found. By streamlining the literature review process, ASK enables researchers to advance scientific discovery.

5. Human-AI Collaboration

The ORKG takes an expert-sourced approach to describe knowledge in a structured form. This means that human intelligence plays a key role. However, we leverage AI to provide user assistance where possible. Broadly, ORKG user interactions can be divided into knowledge *consumers* and knowledge *creators*. AI assistance mainly targets knowledge creators, helping them to 1) simplify tasks, 2) reduce workload, and 3) improve data quality. Creating knowledge includes both generating new knowledge and curating and validating existing knowledge. The main task for ORKG content creators can be described as transforming unstructured scholarly knowledge (i.e., for scholarly articles) to structured scholarly knowledge. This is a multifaceted task that includes both data modeling and knowledge extraction.

5.1. Classification of Human-AI Collaboration

In the ORKG interface, different types of Human-AI interaction take place. Figure 9 displays a classification of these interactions. The AI-Absent approach is a traditional approach where users transform the data manually, without any AI assistance. In the ORKG, it is possible to use the system in such a way that no AI interaction happens. This

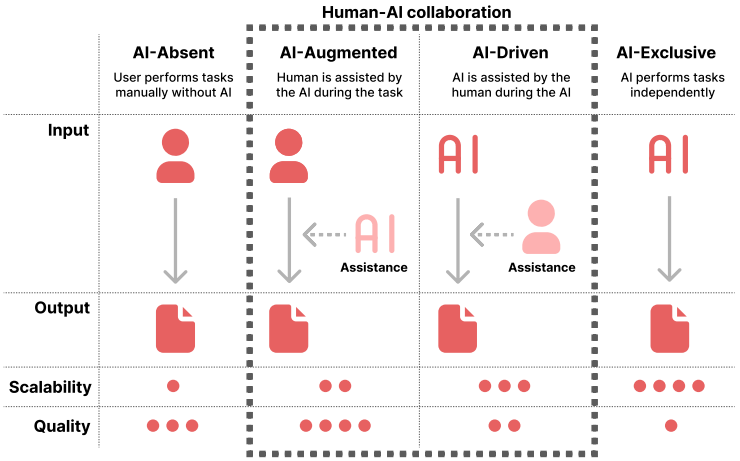

Figure 9. A classification of Human-AI interaction styles, where AI-Augmented and AI-Driven specifically relate to Human-AI collaboration. In these approaches, either the human or the machine provides support during a specific task.

means knowledge can be structured fully in a manual way. Scalability is a challenge with this approach, as you need more manual labor, to transform more knowledge. However, the reliability of the data transformation is generally high compared to the other methods.

Human-AI collaboration takes place in the AI-Augmented approach. As in the AI-Absent approach, the human still takes the initiative of describing the content in a structured form. Concretely, that means that a user decides which papers to describe and how to do this. However, during the process, the user is assisted with AI support. In the ORKG, this type of user interaction underlies numerous features, such as Smart Suggestions, PDF Annotator, and general recommendations in the User Interface (UI). Compared to the fully manual approach, scalability is improving as efficiency improves. Depending on the use case, the reliability could be the highest of all methods, given that the AI can point users towards mistakes and thus help them to improve the output quality.

Another type of Human-AI collaboration takes place in the AI-Driven interaction approach. Here the AI takes the initiative instead of the human. The human helps the AI in specific stages, for example, to verify, improve, and complete data. This means that the AI automatically processes articles and the user at a later point in time can validate the knowledge. In the ORKG, we experimented with the TinyGenius [94] approach, which follows this workflow of automatic extraction and manual verification. This approach is more scalable as machines are able to transform large quantities of scholarly knowledge. However, the reliability, or quality, of the extractions is lower compared to methods where human intelligence is involved.

Finally, the AI-Exclusive approach does not include human intelligence, meaning that no Human-AI collaboration takes place. The scalability of this approach is highest, as no manual labor is required, and scaling will only depend on machine resources. The reliability of such knowledge transformations is lowest compared to the other methods. This approach is currently not implemented in the ORKG, as data quality is an important aim for the ORKG.

5.2. Implementations in the ORKG

We will now discuss implementations of Human-AI collaboration in the ORKG, specifically focusing on AI-Augmented and AI-Driven interactions. To ensure ORKG users are aware of AI support in the ORKG, we use the same recognizable color scheme (green-blue, as displayed in Figure 10) throughout the UI. Additionally, we use the term "Smart" to indicate that a UI component is supported by AI.

ORKG **Smart Suggestions** uses an AI-Augmented interaction, where users are assisted by AI while performing a specific task. They provide a non-intrusive way of integrating (for instance) LLMs into the information transformation task [95]. This is done by placing a small button with a lightbulb icon at multiple locations in the UI (see Figure 10). Smart Suggestions have been integrated for approximately six tasks, including recommendation of properties, value recommendation for specific properties, assessing descriptiveness of metadata, providing guidance for creating reusable property labels, and more. These tasks are identified by considering previous user interaction by ORKG team members and are therefore meant to partially replace human assistance while using the ORKG. By positioning Smart Suggestion as optional guidance for users, existing workflows are not disturbed, simplifying LLM integration into an existing UI. Feedback is collected to ensure the performance of different LLM tasks can be tracked. Depending on the task, relevant context is inserted into the LLM prompt to provide more accurate suggestions.

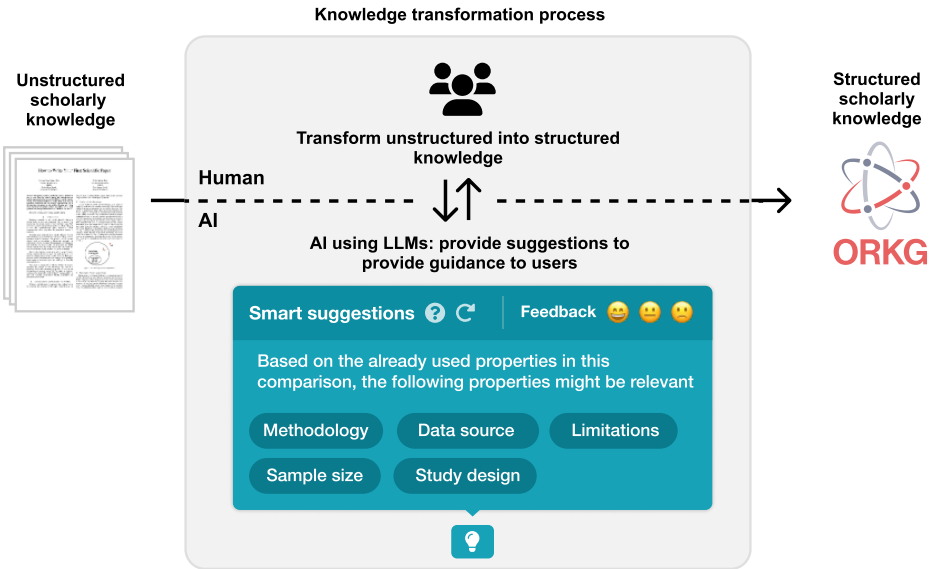

Figure 10. Smart Suggestions as implemented in the ORKG UI displayed by clicking the light bulb button. [96]

The **PDF Annotator** tool enables users to annotate key sentences in a paper with a list of predefined discourse classes [96]. This tool also takes an AI-Augmented approach; while users annotate sentences in PDF articles, they are supported by AI. Key sentences are highlighted in the interface by means of extractive summarization. Users can anno-

tate them by simply clicking on highlighted sentences. When a sentence is selected for annotation, a zero-shot classifier tries to find the set of most relevant classes based on the selected sentence and the complete set of classes.

Finally, we developed an AI-Driven approach by means of crowdsourcing microtasks for validating AI-extracted knowledge [94]. The approach, called TinyGenius, processes the articles offline with a set of NLP tools. Therefore, the AI takes the initiative in this case. Among others, the NLP tasks included Named Entity Recognition (NER), Named Entity Linking (NEL), and summarization. After the data was extracted for a corpus of scientific articles, users were asked to verify the knowledge using microtasks. The task changed depending on which data was being evaluated, but most cases included buttons to indicate that the result was correct, incorrect, or that the evaluator was unsure.

6. Use Cases

This section presents three complementary use cases of the ORKG: 1) an observatory organizing neuro-symbolic AI research according to system composition patterns, 2) agroecological synthesis research, and 3) the organization of food composition information for Cameroon.

6.1. Neuro-symbolic AI Patterns Observatory

The role of an observatory within the ORKG is to organize research in a specific field and ensure a high-quality standard. This involves curating and semantically representing contributions from scientific articles in a manner that is both machine- and human-readable.

Maintaining a Neuro-symbolic AI Patterns Observatory[12] plays an important role in understanding and advancing symbolic and sub-symbolic AI techniques, an area that has garnered significant interest [97]. The observatory, grounded in important ways in the Semantic Web research community, focuses on developing systems that combine Semantic Web and Machine Learning technology, termed Semantic Web and Machine Learning Systems (SWeMLS). It aims to systematically analyze and characterize trends in SWeMLS, addressing the challenges of conducting large-scale, data-driven, and representative studies in this rapidly evolving area.

The observatory's data source is published in [98]. This article conducts a comprehensive systematic mapping study, encompassing information from 470 SWeMLS papers. The main components of this data source are:

- The *SWeMLS ontology* describes the main parts of SWeMLS, including their workflows using boxology patterns. The ontology schema, capturing important SWeMLS characteristics (e.g., StatisticalModel) and relevant instances (e.g., Convolutional neural network), were systematically derived during the scoping and analysis phases of the Systematic Mapping Study (SMS).
- The *SWeMLS-KG knowledge graph* comprises machine-actionable descriptions of almost 500 systems in terms of the SWeMLS ontology.

[12] `https://orkg.org/observatory/Neurosymbolic_artificial_intelligence`

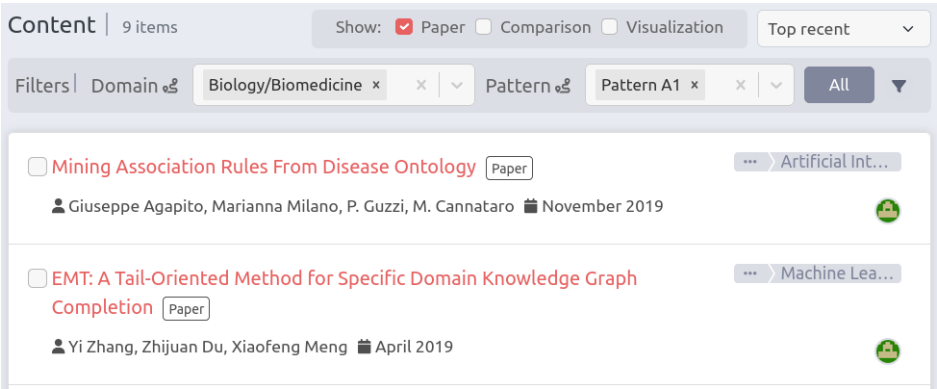

Figure 11. Neuro-symbolic AI Observatory filters.

- The *SWeMLS Pattern Library* contains machine-actionable descriptions of 45 SWeMLS patterns.

This source data was carefully transformed and imported into the ORKG. It is now maintained in the SWeMLS-KG observatory, which offers several functionalities that enhance the utility and impact of the data:

- *Browsing and Visualization:* Users can browse patterns and their instantiations, providing an intuitive understanding of common workflows in SWeMLS.
- *Search and Filter:* The observatory offers robust search and filtering capabilities, allowing users to find contributions based on specific characteristics, such as pattern types or application domains. For example, Figure 11 illustrates the filters on the observatory page, where the user searches for papers that use Pattern A1 in Biology/Biomedicine.
- *Data-Driven Analysis:* Researchers can perform data-driven analyses to uncover trends and best practices within the field of SWeMLS. This is particularly valuable for identifying frequent system patterns used in specific tasks, such as graph completion in the medical domain.
- *FAIR Principles:* The resource adheres to the FAIR principles, ensuring that the data is not only available, but also usable in a variety of research contexts.

To support source data import and new data production, we created ORKG templates[13] (Section 3.2) using the ontology. Users can now add additional related papers to the observatory following the same model and reuse the existing defined patterns. The observatory offers a powerful set of filters to facilitate the accessibility to SWeMLS data, providing researchers with comprehensive insights into the field.

In conclusion, the Neuro-symbolic AI Patterns Observatory in the ORKG demonstrates the potential of neuro-symbolic systems to transform the organization and accessibility of scientific knowledge. By leveraging both Semantic Web and Machine Learning components, this resource offers a platform for understanding and advancing the field of SWeMLS.

[13] `https://orkg.org/template/R575738`

6.2. Agroecological Synthesis Research with the Open Research Knowledge Graph

Synthesis research plays a key role in identifying patterns and processes underlying complex agroecological datasets [99]. Meta-analyses, for instance, are commonly used to integrate findings from multiple studies, generating insights and cross-disciplinary connections that can guide conservation policies and practices [100,101,99]. Despite its usefulness in addressing pressing challenges like climate change and biodiversity loss, conducting synthesis research is increasingly challenging due to the overwhelming volume of academic publications. Even in relatively narrow fields like agroecology, researchers must contend with hundreds of new publications annually [102]. Navigating this publication flood is a major hurdle to staying up-to-date on the latest research and identifying relevant new studies that should be included in a synthesis.

Compounding this problem is the mainstream approach of producing and formally communicating scientific knowledge in unstructured PDF articles [102]. Scientific findings produced in this format are inaccessible to many digital support tools like intelligent search, filter, or other knowledge processing approaches that could make the research life cycle more efficient [1,2]. Rather, most data extraction (from articles) and subsequent database development must be done by hand, which slows the pace and accuracy of research. For example, conducting a meta-analysis or systematic review typically requires researchers to manually extract data from tens or even hundreds of articles and organise this data into a new database, an inaccurate activity that can easily consume three to six months of full-time work [103].

There is a critical need for tools and standards that reduce these points of friction, and facilitate sharing, synthesising, and reproducing complex agroecological research for a range of stakeholders [101]. Here, we briefly present a use case in the ORKG to evaluate how the platform can support these goals in the field of agroecology and outline opportunities for continued development. For a more in-depth discussion of this case study and the ORKG's role in addressing challenges associated with transferring ecological knowledge, we refer readers to Snyder et al. [104].

6.2.1. Research Question

In this use case, we evaluated how the ORKG can support agroecological synthesis research with manually curated data extracted from peer-reviewed articles in the form of an ORKG comparison. Our comparison focused on the yield effects of intercropping cereals and legumes—plants that make atmospheric nitrogen available to other plants—in European agroecosystems. This research question is particularly timely in the context of ongoing efforts to meet global food security needs amidst growing constraints on the availability of arable land [105]. Studies included in the comparison were identified through a Web of Science search, which is described in the ORKG comparison. As our goal was to evaluate the capabilities of the ORKG in scoping articles for a synthesis rather than to conduct a robust synthesis, we did not conduct an exhaustive literature search.

6.2.2. ORKG Comparison

Our interactive comparison [106] can be viewed in full form on the ORKG[14]. We built the comparison using ORKG properties that describe key information published in each

[14] https://doi.org/10.48366/R655553

original contribution such as the study location, experimental methods, quantitative measurements, etc. The process of creating the ORKG comparison was analogous to compiling and coding primary data for a meta-analysis or systematic review. One motivation for organising this data directly in the ORKG rather than in an Excel or CSV file is the ability to leverage ORKG templates. However, as none of the existing templates were suitable for the complex agroecological data we wanted to describe, we had to first develop the required templates. A robust library of templates will be key for attracting and retaining new ORKG users, particularly those unfamiliar with conceptual modelling or unable to perform this task.

Since this comparison exists in a dynamic, open-access environment, other researchers can build upon the search criteria we used to generate the original comparison and integrate additional studies and properties. This dynamic approach to scientific knowledge curation addresses many of the challenges associated with publishing static PDF articles by providing a comprehensive, living resource that can be regularly updated with new and relevant research findings. This is one of the most powerful features of the ORKG.

6.2.3. Visualisations

Once information was coded in the ORKG, we used the ORKG visualisation tool to create a bar chart comparing grain yields across studies (see ORKG comparison). This task was completed in a matter of minutes, which suggests that users with even a low-level of scientific expertise could create meaningful visualisations from existing ORKG comparisons. Comparisons can be visualised in a number of simple formats (e.g., tables, bar charts, etc.) without having to recode the data, allowing users to quickly explore patterns and trends within a dataset.

While ORKG visualisations do not support statistical analyses, they allow researchers to quickly scope trends and patterns. This is particularly useful for systematic mapping in preparation for a review [107]. Moreover, once data is coded into the ORKG, it exists in a machine-reusable format. This means that users can bypass manual data extraction from articles and instead directly download data as a CSV file or import it into a computational environment like R studio. Once in a computational environment, users can use programming languages with enhanced visualisation capabilities (e.g., R or Python) to create custom-made plots and conduct statistical analyses.

6.2.4. Takeaways

Once extracted data are curated in a machine-reusable format in the ORKG, researchers can easily visualise trends and identify knowledge gaps that could be addressed with future research. Given the ease of reusing ORKG data, we can envision a well-populated ORKG drastically accelerating the process of conducting synthesis research. To increase the rigour of ORKG comparisons and their utility as a synthesis scoping tool, a formal literature search must be part of this practice. As in classical meta-analyses and systematic reviews, clearly documenting the literature search process underlying a comparison is a foundational step in creating a robust comparison and should be implemented as standard practice in the ORKG.

Extracting data from PDF articles and organising them in the ORKG was the most time consuming process associated with building our ORKG comparison. Developing

templates to structure data related to common ecology methods and analyses will be key to streamlining this process. Recent ORKG initiatives such as reborn articles [26] or SciKGTeX [23] that allow researchers to produce machine-reusable findings from the outset will also be key in addressing this hurdle. These approaches ensure new research findings can be automatically collected by the ORKG and other knowledge bases, thereby eliminating the need for post-publication data extraction and organisation. Widespread adoption of these tools would facilitate seamless integration of new research into ORKG comparisons that could effectively function as living resources for research, policy, and management decisions.

6.3. Cameroon Food Composition Knowledge Graph

Fair access to food data is an essential catalyst for achieving Sustainable Development Goals, specifically Goal 2 "Zero Hunger", Goal 3 "Ensure healthy lives and promote well-being for all at all ages", and Goal 4 "Quality education". Quality education on food composition is particularly important because it allows people to have a healthy diet and avoid health problems such as malnutrition, obesity, diabetes, some cancers, etc. However, quality education is not a given in developing countries such as Cameroon. Indeed, the most recent versions of the Cameroonian Food Composition Tables (FCT) were published in 1957 [108] and 1966 [109]. These resources are outdated given the current evolution of the country in terms of food habits. However, many Cameroonian foods are described in scientific articles[15]. These resources, mostly in PDF format, are static, not harmonized, difficult to process by machines and difficult to access by different stakeholders.

This section presents how Cameroonian food, their ingredients and their components are being extracted from scientific articles, organized to build the Cameroonian food composition knowledge graph using the neuro-symbolic ORKG [110]. This task is accomplished by an extensive work of semi-automatic collection, extraction and curation of food data stored in scientific articles. Unlike the current FCT which is a static PDF document [109,108], the ORKG provides dynamic resources stored in a KG, which are machine-accessible and easy to continuously update.

The methodology used in this work is based on an iterative and incremental process during which a domain ontology is built, continuously refined, and populated with new food knowledge extracted from scientific articles to construct the food composition KG [111].

6.3.1. TSOTSAGraph: Cameroon Food Composition Knowledge Graph

The Cameroon Food Composition Knowledge Graph (also called TSOTSAGraph for "my food graph" in a local tongue in Cameroon) aims to organize and structure Cameroonian food so as to improve stakeholder access to this knowledge. To construct the KG, one of the most important activities is the development of the food composition ontology (also called TSOTSAOntology - "my food ontology"). This ontology aims to add an expressive layer to food information by introducing the semantic relationships between food, food components, and food ingredients. Thereafter, this ontology is enriched and populated to form the TSOTSAGraph.

[15] `https://orkg.org/list/R693365`

TSOTSAOntology To construct the food composition ontology, extracted information was organized into classes and properties with the help of an expert in food science and nutrition.

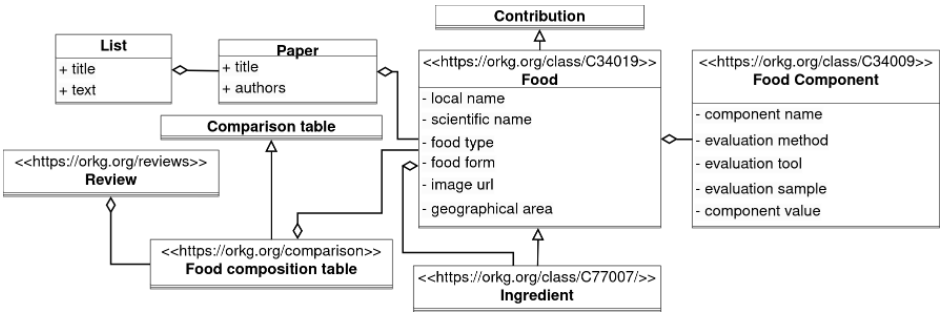

Figure 12. The main classes of the TSOTSAOntology.

The TSOTSAOntology is implemented using the ORKG template editor. This editor allows us to create classes and link these classes to templates (or form) allowing us to describe data properties and object properties. To reuse the large quantity of scientific knowledge already ingested in the ORKG, we employ semantic typing with suggestions based on the entered verbalization of the concept using ChatGPT. Figure 12 presents the main classes.

- *Food, Ingredient*: these classes are used to describe foods and their ingredients;
- *Food component*: this class is used to describe the different components (iron, fat, carbohydrate, etc.) contained in foods;
- *Food Composition Tables*: this class inherits the ORKG comparison table resource. It aims to compare the different ingredients (or) components contained in foods. It is generally associated with a visualization presenting, for instance, which ingredient or component is in small/large quantities in a food;
- Review: the class review aims to provide a food review described as food composition tables.

The *Food*, *Food ingredient* and *Food component* classes are linked to food, food ingredient, and food component templates respectively. They are used as conceptual models for populating the TSOTSAOntology with food data extracted from scientific articles.

TSOTSAGraph Stored in the ORKG, the TSOTSAGraph is obtained by populating the TSOTSAOntology with instances. To this end, the ORKG provides a wizard for adding and annotating papers using templates. Using the paper DOI or title, the ORKG automatically fetches the paper metadata and allows us to identify and import the corresponding template. Language Models are also used during this activity for resource suggestion and recommendation. To easily navigate the TSOTSAGraph, one may use food, food composition, food composition tables and reviews:

- *Food and food composition instances:* we use the ORKG contribution editor to describe food and their components. Each food described in a paper corresponds to one research contribution. More than 500 instances are already ingested in the TSOTSAGraph.

- *Food composition tables instances*[16] correspond to the different ORKG comparisons tables used to compare food, food ingredients, and food components. So far, we have created ten ORKG comparisons.
- *Food composition review:* so far, we have published one review of Cameroon food composition table[17]. This review will be updated as more food data are stored and organized and more food composition tables are created.

7. Conclusions and Future Work

The Open Research Knowledge Graph (ORKG) represents a transformative approach to scholarly communication, making scientific knowledge more accessible, discoverable, and actionable. By leveraging the power of neuro-symbolic technologies and providing robust tools for content production, curation, and use, the ORKG is poised to shape the future of research and knowledge management.

We see the current state of the ORKG as described in this chapter only as the first step in a larger research and development agenda. Our long-term aim is to develop an advanced neuro-symbolic AI system designed to support all phases of scientific research. Building on the ORKG, a Neuro-Symbolic Research Co-Pilot can assist researchers in hypothesis generation, literature review, experimental design, data analysis, and dissemination of findings. The ultimate goal is to create a versatile, intelligent assistant that augments researchers' capabilities, accelerates scientific discovery, and fosters interdisciplinary collaboration. One critical area in this journey is the expansion and refinement of semantic templates to cover a broader range of research domains, ensuring a comprehensive and accurate representation of diverse scientific contributions. Additionally, integrating more advanced neuro-symbolic AI techniques (e.g. models fine-tuned for the scholarly knowledge organization tasks) will improve the system's ability to process and understand complex scientific texts, leading to even richer and more precise knowledge graphs. Another promising direction involves enhancing user interfaces and developing more intuitive tools for researchers to interact with the ORKG, including advanced visualizations, dashboards and natural language assistance capabilities. Expanding collaborations with academic publishers, research institutions, funders, and other stakeholders will also be essential to drive widespread adoption and contribute to the continuous curation and updating of the knowledge graph. Lastly, efforts to ensure the ethical use and transparency of the ORKG, addressing issues such as data privacy and bias, will be paramount in maintaining trust and integrity within the scientific community. By pursuing these future directions, the ORKG aims to solidify its role as a cornerstone of the next generation of scholarly communication.

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[16] Example: `https://orkg.org/comparison/R689520`
[17] `https://orkg.org/review/R700894`

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Data and Code Availability

The ORKG promotes open science and transparency. The codebase is open source, MIT licensed, and made available to the community at gitlab.com/TIBHannover/orkg. The data is licensed under CC0 1.0 Universal except data sourced from named third-party sources, which are republished with permission under the original license. The license statements are available at orkg.org/page/license.

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